

Technology Stack

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / ClientSide	HTML5, CSS3, JavaScript	Used to create a responsive and userfriendly interface for entering applicant details, validating inputs, and displaying eligibility prediction results.
2	Backend / ServerSide	Python, Flask	Flask is used to develop the web application, handle routing, process user inputs, load the trained machine learning model, and return prediction results.
3	Data Storage / Machine Learning	CSV Dataset, Pandas DataFrames, Joblib (.pkl Files)	The CSV dataset is used for training the model, while Joblib stores the trained model, scaler, encoder, and feature columns for real-time predictions.
4	Cloud / Hosting / Deployment	Render, Unicorn	Render is used to deploy the Flask application on the cloud, while Unicorn serves as the production WSGI server for handling requests efficiently.

5	Version Control & Collaboration	Git, GitHub	Used for source code management, version control, collaboration among team members, and deployment integration.
6	Machine Learning Libraries & Development Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, OpenPyXL, Joblib	Used for data preprocessing, feature engineering, model training, evaluation, visualization, model serialization, and Excel dataset support.